

The Downward Closure Property of Frequent Patterns

- ❑ Observation: From TDB₁: $T_1: \{a_1, \dots, a_{50}\}$; $T_2: \{a_1, \dots, a_{100}\}$
 - ❑ We get a frequent itemset: $\{a_1, \dots, a_{50}\}$
 - ❑ Also, its subsets are all frequent: $\{a_1\}, \{a_2\}, \dots, \{a_{50}\}, \{a_1, a_2\}, \dots, \{a_1, \dots, a_{49}\}, \dots$
 - ❑ There are some hidden relationships among frequent patterns!
- ❑ The **downward closure (also called “Apriori”)** property of frequent patterns
 - ❑ If **{beer, diaper, nuts}** is frequent, so is **{beer, diaper}**
 - ❑ Every transaction containing {beer, diaper, nuts} also contains {beer, diaper}
 - ❑ Apriori: Any subset of a frequent itemset must be frequent
- ❑ Efficient mining methodology
 - ❑ If **any subset of an itemset S** is infrequent, then there is no chance for S to be frequent—why do we even have to consider S!?  **A sharp knife for pruning!**

Apriori Pruning and Scalable Mining Methods

- ❑ Apriori pruning principle: If there is any itemset which is infrequent, its superset should not even be generated! (Agrawal & Srikant @VLDB'94, Mannila, et al. @ KDD' 94)
- ❑ Scalable mining Methods: Three major approaches
 - ❑ Level-wise, join-based approach: Apriori (Agrawal & Srikant@VLDB'94)
 - ❑ Vertical data format approach: Eclat (Zaki, Parthasarathy, Ogihara, Li @KDD'97)
 - ❑ Frequent pattern projection and growth: FPgrowth (Han, Pei, Yin @SIGMOD'00)

Apriori: A Candidate Generation & Test Approach

- ❑ Outline of Apriori (level-wise, candidate generation and test)
 - ❑ **Scan** DB once to get frequent 1-itemset
 - ❑ **Repeat**
 - ❑ Generate length-($k+1$) candidate itemsets from length- k frequent itemsets
 - ❑ Test the candidates against DB to find frequent ($k+1$)-itemsets
 - ❑ Set $k := k + 1$
 - ❑ **Until** no frequent or candidate set can be generated
 - ❑ Return all the frequent itemsets derived